# DBMS LAB 7


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- 123CS0061


# Assignment 1

# Step 1

```oracle
create table students(
    student_id number primary key,
    name varchar2(50),
    age number,
    dept varchar2(50),
    marks number
);

create table courses(
    course_id number primary key,
    course_name varchar2(50),
    dept varchar2(50),
    credits number
);

select * from students;
select * from courses;
```

Script Output | Query Result | Query Result 1 | Query Result 2

SQL | All Rows Fetched: 0 in 0.004 seconds

| COURSE_ID | COURSE_... | DEPT | CREDITS |
|---|---|---|---|

```oracle
insert into students (student_id, name, age, dept, marks) values (1, 'alice', 19, 'cse', 85);
insert into students (student_id, name, age, dept, marks) values (2, 'bob', 20, 'cse', 70);
insert into students (student_id, name, age, dept, marks) values (3, 'carol', 19, 'ece', 92);
insert into students (student_id, name, age, dept, marks) values (4, 'david', 21, 'ece', 60);
insert into students (student_id, name, age, dept, marks) values (5, 'eve', 20, 'mech', 75);

select * from students;
select * from courses;
```

pt Output | Query Result | Query Result 1 | Query Result 2

SQL | All Rows Fetched: 5 in 0.005 seconds

| | STUDENT_ID | NAME | AGE | DEPT | MARKS |
|---|---|---|---|---|---|
| 1 | 1 | alice | 19 | cse | 85 |
| 2 | 2 | bob | 20 | cse | 70 |
| 3 | 3 | carol | 19 | ece | 92 |
| 4 | 4 | david | 21 | ece | 60 |
| 5 | 5 | eve | 20 | mech | 75 |

```oracle
insert into courses (course_id, course_name, dept, credits) values (101, 'dbms', 'cse', 4);
insert into courses (course_id, course_name, dept, credits) values (102, 'networks', 'cse', 3);
insert into courses (course_id, course_name, dept, credits) values (103, 'vlsi', 'ece', 4);
insert into courses (course_id, course_name, dept, credits) values (104, 'signals', 'ece', 3);
insert into courses (course_id, course_name, dept, credits) values (105, 'thermo', 'mech', 4);
```

cript Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3

SQL | All Rows Fetched: 5 in 0.002 seconds

| | COURSE_ID | COURSE_NAME | DEPT | CREDITS |
|---|---|---|---|---|
| 1 | 101 | dbms | cse | 4 |
| 2 | 102 | networks | cse | 3 |
| 3 | 103 | vlsi | ece | 4 |
| 4 | 104 | signals | ece | 3 |
| 5 | 105 | thermo | mech | 4 |

# Step 3

## Part A: Basic SubQueries

```sql
select *
from students
where marks > (
    select avg(marks)
    from students
);
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3

SQL | All Rows Fetched: 2 in 0.004 seconds

| | STUDENT_ID | NAME | AGE | DEPT | MARKS |
|---|---|---|---|---|---|
| 1 | 1 | alice | 19 | cse | 85 |
| 2 | 3 | carol | 19 | ece | 92 |

```sql
select *
from students
where marks = (
    select max(marks)
    from students
);
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3

SQL | All Rows Fetched: 1 in 0.003 seconds

| | STUDENT_ID | NAME | AGE | DEPT | MARKS |
|---|---|---|---|---|---|
| 1 | 3 | carol | 19 | ece | 92 |

```sql
select *
from students
where dept = (
    select dept
    from students
    where name = 'alice'
);
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3

SQL | All Rows Fetched: 2 in 0.005 seconds

| | STUDENT_ID | NAME | AGE | DEPT | MARKS |
|---|---|---|---|---|---|
| 1 | 1 | alice | 19 | cse | 85 |
| 2 | 2 | bob | 20 | cse | 70 |

# Part B: Multi-row Subqueries

```sql
select *
from students
where dept in (
    select dept
    from courses
    where credits = 4
);

select * from students;
select * from courses;
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3

SQL | All Rows Fetched: 5 in 0.005 seconds

| | STUDENT_ID | NAME | AGE | DEPT | MARKS |
|---|---|---|---|---|---|
| 1 | 1 | alice | 19 | cse | 85 |
| 2 | 2 | bob | 20 | cse | 70 |
| 3 | 3 | carol | 19 | ece | 92 |
| 4 | 4 | david | 21 | ece | 60 |
| 5 | 5 | eve | 20 | mech | 75 |

```sql
select *
from students
where marks > (
    select min(marks)
    from students
    where dept = 'cse'
);

select * from students;
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3

SQL | All Rows Fetched: 3 in 0.003 seconds

| | STUDENT_ID | NAME | AGE | DEPT | MARKS |
|---|---|---|---|---|---|
| 1 | 1 | alice | 19 | cse | 85 |
| 2 | 3 | carol | 19 | ece | 92 |
| 3 | 5 | eve | 20 | mech | 75 |

# Part C: Correlated Subqueries

```sql
select *
from students s
where marks >= (
    select avg(marks)
    from students x
    where x.dept = s.dept
);

select * from students;
select * from courses;
```

Query Result 3 — All Rows Fetched: 3 in 0.005 seconds

| | STUDENT_ID | NAME | AGE | DEPT | MARKS |
|---|---|---|---|---|---|
| 1 | 1 | alice | 19 | cse | 85 |
| 2 | 3 | carol | 19 | ece | 92 |
| 3 | 5 | eve | 20 | mech | 75 |

```sql
select *
from students s
where s.age > (
    select min(age)
    from students x
    where x.dept = s.dept
);

select * from students;
select * from courses;
```

Query Result 3 — All Rows Fetched: 2 in 0.007 seconds

| | STUDENT_ID | NAME | AGE | DEPT | MARKS |
|---|---|---|---|---|---|
| 1 | 2 | bob | 20 | cse | 70 |
| 2 | 4 | david | 21 | ece | 60 |

# Part D: Placement Subqueries

```sql
select *
from (
    select student_id as stid, name, age, dept, marks + 5 as new_marks
    from students
) st_temp;

select * from students;
select * from courses;
```

Query Result 3 — All Rows Fetched: 5 in 0.002 seconds

| | STID | NAME | AGE | DEPT | NEW_MARKS |
|---|---|---|---|---|---|
| 1 | 1 | alice | 19 | cse | 90 |
| 2 | 2 | bob | 20 | cse | 75 |
| 3 | 3 | carol | 19 | ece | 97 |
| 4 | 4 | david | 21 | ece | 65 |
| 5 | 5 | eve | 20 | mech | 80 |

```sql
select dept
from students
group by dept
having avg(marks) > (
    select avg(marks)
    from students
);
```

Query Result 3 — All Rows Fetched: 1 in 0.001 seconds

| | DEPT |
|---|---|
| 1 | cse |

```sql
select student_id, name, age, dept, marks, (
    select avg(marks) from students
) as avg_marks
from students;
```

Query Result 3 — All Rows Fetched: 5 in 0.002 seconds

| | STUDENT_ID | NAME | AGE | DEPT | MARKS | AVG_MARKS |
|---|---|---|---|---|---|---|
| 1 | 1 | alice | 19 | cse | 85 | 76.4 |
| 2 | 2 | bob | 20 | cse | 70 | 76.4 |
| 3 | 3 | carol | 19 | ece | 92 | 76.4 |
| 4 | 4 | david | 21 | ece | 60 | 76.4 |
| 5 | 5 | eve | 20 | mech | 75 | 76.4 |

# Part E: Placement Subqueries

```sql
update students
set marks = marks + 10
where marks < (
    select avg(marks) from students
);

select * from students;
select * from courses;
```

Script Output × | Query Result × | Query Result 1 × | Query Result 2 × | Query Result 3 ×

SQL | All Rows Fetched: 5 in 0.001 seconds

| | STUDENT_ID | NAME | AGE | DEPT | MARKS |
|---|---|---|---|---|---|
| 1 | 1 | alice | 19 | cse | 85 |
| 2 | 2 | bob | 20 | cse | 80 |
| 3 | 3 | carol | 19 | ece | 92 |
| 4 | 4 | david | 21 | ece | 70 |
| 5 | 5 | eve | 20 | mech | 85 |

```sql
delete from students
where dept in (
    select dept
    from students
    minus
    select dept
    from courses
    where credits = 4
);
```

# Assignment 2

# Step 1

```sql
insert into books (book_id, title, author, price, stock) values (1, 'dbms made easy', 'navathe', 500, 20);
insert into books (book_id, title, author, price, stock) values (2, 'learning sql', 'alan beaulieu', 400, 15);
insert into books (book_id, title, author, price, stock) values (3, 'operating system', 'galvin', 600, 10);
insert into books (book_id, title, author, price, stock) values (4, 'networks basics', 'tanenbaum', 450, 12);
insert into books (book_id, title, author, price, stock) values (5, 'python coding', 'zed shaw', 350, 25);

select * from books;
select * from orders;
```

Script Output × | Query Result × | Query Result 1 × | Query Result 2 × | Query Result 3 × | Query Result 4 × | Query Result 5 ×

SQL | All Rows Fetched: 5 in 0.002 seconds

| | BOOK_ID | TITLE | AUTHOR | PRICE | STOCK |
|---|---|---|---|---|---|
| 1 | 1 | dbms made easy | navathe | 500 | 20 |
| 2 | 2 | learning sql | alan beaulieu | 400 | 15 |
| 3 | 3 | operating system | galvin | 600 | 10 |
| 4 | 4 | networks basics | tanenbaum | 450 | 12 |
| 5 | 5 | python coding | zed shaw | 350 | 25 |

```sql
insert into orders (order_id, book_id, quantity, customer) values (101, 1, 2, 'alice');
insert into orders (order_id, book_id, quantity, customer) values (102, 2, 1, 'bob');
insert into orders (order_id, book_id, quantity, customer) values (103, 3, 3, 'carol');
insert into orders (order_id, book_id, quantity, customer) values (104, 5, 2, 'david');
insert into orders (order_id, book_id, quantity, customer) values (105, 2, 4, 'eve');
```

Script Output × | Query Result × | Query Result 1 × | Query Result 2 × | Query Result 3 × | Query Result 4 × | Query Result 5 × | Query Result 6 ×

SQL | All Rows Fetched: 5 in 0.003 seconds

| | ORDER_ID | BOOK_ID | QUANTITY | CUSTOMER |
|---|---|---|---|---|
| 1 | 101 | 1 | 2 | alice |
| 2 | 102 | 2 | 1 | bob |
| 3 | 103 | 3 | 3 | carol |
| 4 | 104 | 5 | 2 | david |
| 5 | 105 | 2 | 4 | eve |

# Step 3

## Part A: Basic SubQueries

```
select *
from books
where price = (
    select max(price) from books
);

select * from books;
select * from orders;
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3 | Query Result 4 | Query Result 5 | Query Result 6

SQL | All Rows Fetched: 1 in 0.001 seconds

| | BOOK_ID | TITLE | AUTHOR | PRICE | STOCK |
|---|---|---|---|---|---|
| 1 | 3 | operating system | galvin | 600 | 10 |

```
select distinct customer
from orders o
where (
    select price
    from books b
    where b.book_id = o.book_id
) < (
    select avg(price) from books
);

select * from books;
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3 | Query Result 4 | Query Result 5 | Query Result 6

SQL | All Rows Fetched: 3 in 0.003 seconds

| | CUSTOMER |
|---|---|
| 1 | bob |
| 2 | david |
| 3 | eve |

```
select *
from books
where author = (
    select author
    from books
    where title = 'dbms made easy'
);
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3 | Query Result 4 | Query Result 5 | Query Result 6

SQL | All Rows Fetched: 1 in 0.003 seconds

| | BOOK_ID | TITLE | AUTHOR | PRICE | STOCK |
|---|---|---|---|---|---|
| 1 | 1 | dbms made easy | navathe | 500 | 20 |

# Part B: Multi-row Subqueries

```
select *
from books b
where b.price > (
    select max(price)
    from books x
    where x.book_id = (
        select book_id
        from orders o
        where o.customer = 'alice'
    )
);

select * from books;
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3 | Query Result 4 | Query Result 5 | Query Result 6

SQL | All Rows Fetched: 1 in 0.005 seconds

| | BOOK_ID | TITLE | AUTHOR | PRICE | STOCK |
|---|---|---|---|---|---|
| 1 | 3 | operating system | galvin | 600 | 10 |

```
select customer
from orders o
where o.book_id in (
    select book_id
    from orders x
    where x.customer = 'eve'
);

select * from books;
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3 | Query Result 4 | Query Result 5 | Query Result 6

SQL | All Rows Fetched: 2 in 0.004 seconds

| | CUSTOMER |
|---|---|
| 1 | bob |
| 2 | eve |

# Part C: Correlated Subqueries

```
select customer
from orders o
where o.quantity > (
    select avg(quantity)
    from orders x
    where o.book_id = x.book_id
);
```

Script Output × | Query Result × | Query Result 1 × | Query Result 2 × | Query Result 3 × | Query Result 4 × | Query Result 5 × | Query Result 6 ×

SQL | All Rows Fetched: 1 in 0.006 seconds

| | CUSTOMER |
|---|---|
| 1 | eve |

```
select book_id, title
from books b
where stock < (
    select sum(quantity)
    from orders o
    where o.book_id = b.book_id
);

select * from books;
```

Script Output × | Query Result × | Query Result 1 × | Query Result 2 × | Query Result 3 × | Query Result 4 × | Query Result 5 × | Query Result 6 ×

SQL | All Rows Fetched: 0 in 0.005 seconds

| BOOK_ID | TITLE |
|---|---|

# Part D: Placement Subqueries

```
select *
from (
    select book_id,
           title,
           author,
           price,
           stock,
           price * (
                select sum(quantity)
                from orders o
                where o.book_id = b.book_id
           ) as total_revenue
    from books b
) books_total;
```

Script Output × | Query Result × | Query Result 1 × | Query Result 2 × | Query Result 3 × | Query Result 4 × | Query Result 5 × | Query Result 6 ×

SQL | All Rows Fetched: 5 in 0.001 seconds

| | BOOK_ID | TITLE | AUTHOR | PRICE | STOCK | TOTAL_REVENUE |
|---|---|---|---|---|---|---|
| 1 | 1 | dbms made easy | navathe | 500 | 20 | 1000 |
| 2 | 2 | learning sql | alan beaulieu | 400 | 15 | 2000 |
| 3 | 3 | operating system | galvin | 600 | 10 | 1800 |
| 4 | 5 | python coding | zed shaw | 350 | 25 | 700 |
| 5 | 4 | networks basics | tanenbaum | 450 | 12 | (null) |

```
select author
from books
group by author
having avg(price) > (
    select avg(price) from books
);

select * from books;
```

Script Output × | Query Result × | Query Result 1 × | Query Result 2 × | Query Result 3 × | Query Result 4 × | Query Result 5 × | Query Result 6 ×

SQL | All Rows Fetched: 2 in 0 seconds

| | AUTHOR |
|---|---|
| 1 | navathe |
| 2 | galvin |

```
select order_id, book_id, quantity, customer,
    (
        select title
        from books b
        where b.price >= (
            select max(price) from books
        )
    ) as highest_price_book
from orders;

select * from books;
```

Script Output × | Query Result × | Query Result 1 × | Query Result 2 × | Query Result 3 × | Query Result 4 × | Query Result 5 × | Query Result 6 ×

SQL | All Rows Fetched: 5 in 0.003 seconds

| | ORDER_ID | BOOK_ID | QUANTITY | CUSTOMER | HIGHEST_PRICE_BOOK |
|---|---|---|---|---|---|
| 1 | 101 | 1 | 2 | alice | operating system |
| 2 | 102 | 2 | 1 | bob | operating system |
| 3 | 103 | 3 | 3 | carol | operating system |
| 4 | 104 | 5 | 2 | david | operating system |
| 5 | 105 | 2 | 4 | eve | operating system |

# Part E: Placement Subqueries

```sql
update books
set price = price + 50
where price < (
    select avg(price) from books
);

select * from books;
select * from orders;
```

cript Output × | Query Result × | Query Result 1 × | Query Result 2 × | Query Result 3 × | Query Result 4 × | Query Result 5 × | Query Result 6 ×

SQL | All Rows Fetched: 5 in 0.001 seconds

| | BOOK_ID | TITLE | AUTHOR | PRICE | STOCK |
|---|---|---|---|---|---|
| 1 | 1 | dbms made easy | navathe | 500 | 20 |
| 2 | 2 | learning sql | alan beaulieu | 450 | 15 |
| 3 | 3 | operating system | galvin | 600 | 10 |
| 4 | 4 | networks basics | tanenbaum | 500 | 12 |
| 5 | 5 | python coding | zed shaw | 400 | 25 |

```sql
delete from books b
where (
    b.book_id not in (
        select book_id from orders
    )
);

select * from books;
select * from orders;
```

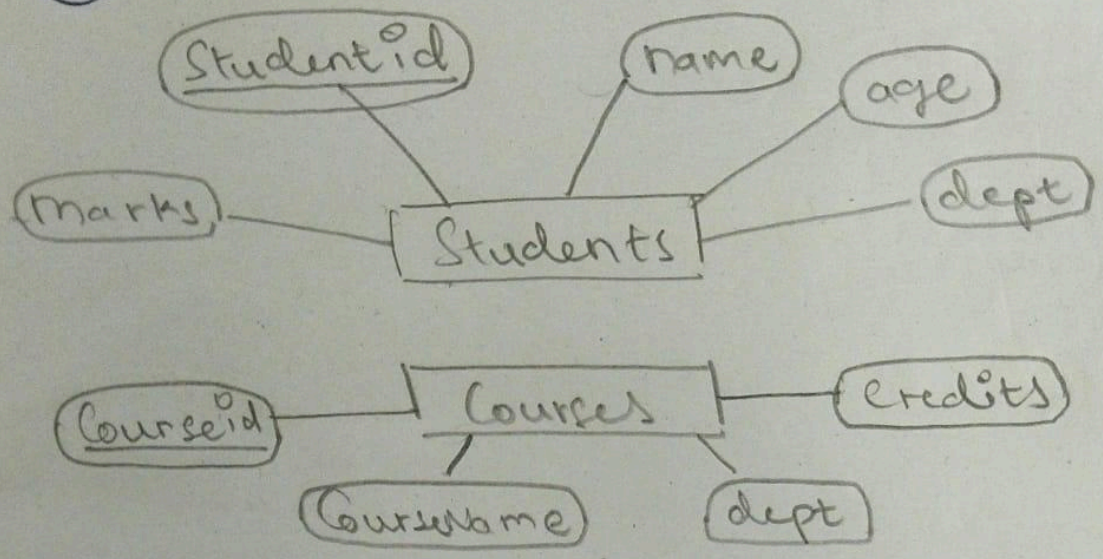
cript Output × | Query Result × | Query Result 1 × | Query Result 2 × | Query Result 3 × | Query Result 4 × | Query Result 5 × | Query Result 6 × | Query Result 7 ×

SQL | All Rows Fetched: 4 in 0.001 seconds

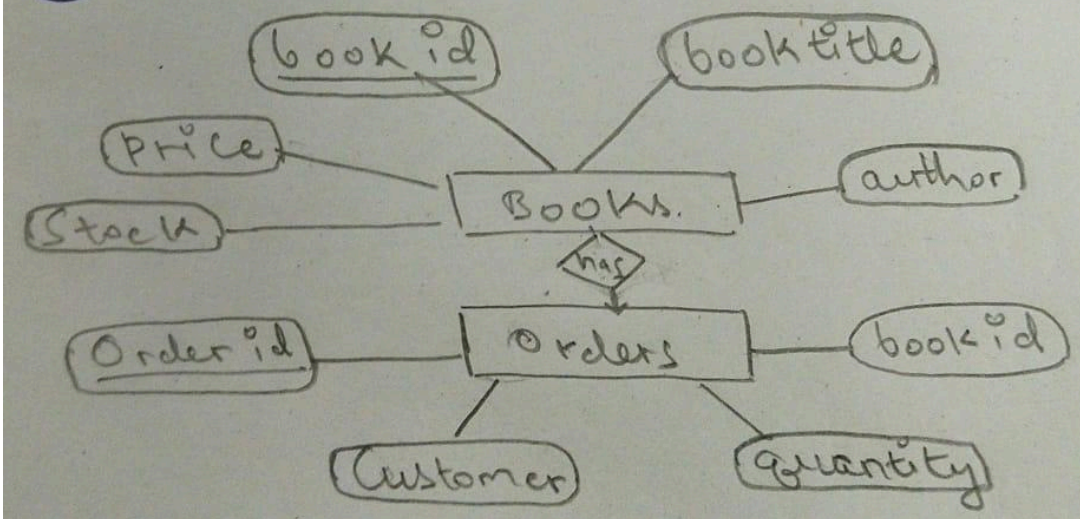
| | BOOK_ID | TITLE | AUTHOR | PRICE | STOCK |
|---|---|---|---|---|---|
| 1 | 1 | dbms made easy | navathe | 500 | 20 |
| 2 | 2 | learning sql | alan beaulieu | 450 | 15 |
| 3 | 3 | operating system | galvin | 600 | 10 |
| 4 | 5 | python coding | zed shaw | 400 | 25 |

# Lab 7

(A)


Studentid
name
age
marks
Students
dept
Courseid
Courses
credits
Coursename
dept

(B)


bookid
booktitle
price
Books
author
Stock
has
Orderid
Orders
bookid
Customer
quantity